Task 1: A Course List with Detail Screens

```
import Foundation
struct Course: Identifiable, Hashable {
    let id: Int
    let title: String
    let hours: Int
}

let courses = [
    Course(id: 1, title: "Swift Basics", hours: 10),
    Course(id: 2, title: "SwiftUI", hours: 15),
    Course(id: 3, title: "iOS Development", hours: 20)
]
func showCourseDetails(_ course: Course) {
    print("\nCourse Details")
    print("Title: \(course.title)")
    print("Hours: \(course.hours)")
}
print("Course List")
for course in courses {
    print("\(course.id). \(course.title)")
}
showCourseDetails(courses[0])
showCourseDetails(courses[2])
```



Task 2: Reusable Rating View and Card Modifier

```
import Foundation
struct RatingView {
    var rating: Int
    let totalStars = 5
    mutating func setRating(_ value: Int) {
        rating = max(0, min(value, totalStars))
    }

    func display() {
        let stars = String(repeating: "★", count: rating)
        let emptyStars = String(repeating: "☆", count: totalStars - rating)

        print("Rating: \(stars)\(emptyStars) (\(rating) of \(totalStars))")
```

```
    }
}
var rating = RatingView(rating: 3)
print("Initial rating:")
rating.display()
rating.setRating(5)
print("\nAfter selecting five stars:")
rating.display()
print("\nCard Style")
print("Padding: 16")
print("Background: White")
print("Corner radius: 12")
print("Shadow: Enabled")
```

Output

```
Initial rating:
Rating: ★★★☆☆ (3 of 5)

After selecting five stars:
Rating: ★★★★★ (5 of 5)

Card Style
Padding: 16
Background: White
Corner radius: 12
Shadow: Enabled
```

Task 3: Programmatic Navigation and Tabs

```
import Foundation
var navigationStack = [1]

func goToNext() {
    if let current = navigationStack.last {
        navigationStack.append(current + 1)
    }
    print("Current screen: \(navigationStack.last!)")
}

func goBackToRoot() {
    navigationStack = [1]
    print("Returned to root screen")
    print("Current screen: \(navigationStack.last!)")
}

print("Home screen: 1")

print("\nGo to next screen")
goToNext()

print("\nGo to next screen")
goToNext()
```

```
print("\nGo back to root")
goBackToRoot()

print("\nTab 1: Navigation")
print("Tab 2: Settings")
```

Output

```
Home screen: 1

Go to next screen
Current screen: 2

Go to next screen
Current screen: 3

Go back to root
Returned to root screen
Current screen: 1

Tab 1: Navigation
Tab 2: Settings
```